

Xavier Kervyn

Max Planck Institute for Physics, Boltzmannstraße 8, 85748 Garching bei München, Germany
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CURRENT RESEARCH INTERESTS

- ★ **String theory:** flat-space and AdS string amplitudes, double copy, motives, tensionless strings, surfaceology
- ★ **Top-down holography:** flat-space and twisted holography, AdS/CFT, decoupling limits of string/M-theory
- ★ **Mathematics:** algebraic geometry, resurgence theory, intersection theory, celestial and Carrollian symmetries

ACADEMIC EMPLOYMENT

- Max Planck Institute for Physics** Garching bei München, Munich, Germany 10/2025
Doctoral Researcher, MPP QFT + String Theory Groups –present
- International Max Planck Research School (**IMPRS**) PhD candidate
 - Advised by [Stephan Stieberger](#) (string theory) and [Johannes Henn](#) (quantum field theory)
- Max Planck Institute for Physics** Garching bei München, Munich, Germany & remote 09/2024
Graduate Researcher, MPP QFT + String Theory Groups 09/2025
- Master's thesis on string amplitudes and flat-space holography
 - Advised by [Stephan Stieberger](#) (string theory)

EDUCATION

- Ludwig-Maximilians-Universität München** Munich, Germany 10/2025
PhD in Theoretical Physics –present
- Advanced coursework in mathematics: algebraic geometry, resurgence theory
- Ludwig-Maximilians- & Technische Universität München** Munich, Germany 10/2023
Master of Science in Theoretical and Mathematical Physics (**TMP**, with Distinction) 09/2025
- Thesis: *Towards a String Worldsheet Description of Celestial Holography* (1.0, High Distinction, 1 paper)
 - Awarded 2 scholarships. Research at **Niels Bohr Institute & Max Planck Institute for Physics**
- Peterhouse, University of Cambridge** Cambridge, United Kingdom 10/2022
Master of Advanced Study in Applied Mathematics (**Part III**, with First Class Honours) 06/2023
- Part III essay: *BMS Symmetries of Gravitational Scattering* (86%, Distinction, published as review)
 - Awarded 3 scholarships and 3 prizes. Summer research at **Università degli Studi di Padova**
- Ecole Polytechnique Fédérale de Lausanne** Lausanne, Switzerland 09/2019
Bachelor of Science in Physics 08/2022
- Awarded 2 fellowships and 1 scholarship. Research at **CERN & ETH Zürich**

RESEARCH STAYS & LONG-TERM ACADEMIC VISITS

- Niels Bohr Institute** Copenhagen, Denmark 09/2024
Visiting Student, Theoretical High-Energy Physics 01/2025
- Study project on the superconformal bootstrap of gauge theories with defects
 - Supervised by [Charlotte Kristjansen](#) and [Adam Chalabi](#)
- Università degli Studi di Padova** Padova, Italy 07/2023
Visitor, Theoretical Physics
- Research on integrability in superconformal field theories and AdS/CFT
 - Supervised by [Alessandro Sfondrini](#) and [Davide Polvara](#)

CERN CMS collaboration Meyrin, Switzerland	06/2022
Undergraduate Researcher, ETH Zürich High-Energy Physics group	07/2022
– Wrote a Python package for the new high-luminosity CMS electrocalorimeter, detector testing	
ETH Zürich Zürich, Switzerland	09/2021
Undergraduate Researcher, Exoplanets and Habitability group	12/2021
– Experimental concept of the LIFE mission, modelled imperfect nulling in the interferometer	

SELECTED AWARDS, GRANTS & SCHOLARSHIPS

Bourse d'Études – <i>Fondation du Domaine de Villette</i> // CHF 5'000	10/2024
Erasmus+ Scholarship – <i>Ludwig-Maximilians-Universität München</i> // €3'000	05/2024
College Prize in Applied Mathematics – <i>Peterhouse, University of Cambridge</i> // £300	12/2023
Retrospective Scholarship in Mathematics – <i>Peterhouse, University of Cambridge</i> // £125	12/2023
Bourse d'Études – <i>Fondation du Domaine de Villette</i> // CHF 5'000	06/2023
Greta Burkill Fund Award – <i>Peterhouse, University of Cambridge</i> // £300	03/2023
Bruckmann Fund Award – <i>Peterhouse, University of Cambridge</i> // £300	03/2023
Annual Scholarship – <i>Swiss Study Foundation</i> // CHF 20'000	11/2022
Excellence Scholarship – <i>Colbianco Stiftung</i> // CHF 2'000	08/2022
Fellowship – <i>Swiss Study Foundation</i> & <i>e-fellows.net</i>	05/2022
Swiss Mobility Program Grant – <i>Ecole Polytechnique Fédérale de Lausanne</i> // CHF 1'500	09/2021
Baccalaureate Merit Award – <i>Région Provence-Alpes Côte d'Azur</i> // €400	07/2019

PUBLICATIONS & PREPRINTS ([Inspire HEP](#); *:under review) (author ordering is alphabetical in hep-th)

Articles:

3. X. Kervyn and S. Stieberger, *Resurgence of high-energy string amplitudes*, [2604.07444](#).
2. X. Kervyn and S. Stieberger, *High-energy string theory and the celestial sphere*, *Journal of High Energy Physics* **09** (2025) 044 [[2504.13738](#)]. (based on my master thesis at LMU/TUM)
1. X. Kervyn, D. Polvara and A. Sfondrini, *Thermodynamics of integrable $\mathcal{N} = 2$ theories, squared*, *Journal of High Energy Physics* **09** (2025) 018 [[2502.10356](#)].

Reviews:

1. X. Kervyn, *BMS symmetries of gravitational scattering*, *Nuclear Physics B* **1017** (2025) 116948 [[2308.12979](#)]. (based on my Part III essay at Cambridge)

PARTICIPATION TO CONFERENCES (*: scheduled, P: poster, T: invited talk, G: gong show)

– Fifty Years of BRST, A. Sommerfeld Center for Theoretical Physics, Munich, Germany [link]	03/2026
– Symbology @15, MPI MPP, Munich, Germany [link]	12/2025
– Feynman Calculus and its Applications to Gravity and Particle Physics [link , <i>online</i>]	11/2025
– DFG meeting: Modern Foundations of Scattering Amplitudes, BCTP, Bonn, Germany [link]	11/2025
– From Good Cuts to Celestial Holography, St Anthony's College, Oxford University, UK [link]	07/2025
– Infrared Surprises of Scattering Amplitudes, CERN, Meyrin, Switzerland [link , <i>online</i>]	05/2025
– Strings 2024, CERN, Meyrin, Switzerland [link]	06/2024
– Carrollian Physics and Holography, ESI, Vienna, Austria [link]	04/2024
– Workshop [...] in Analysis and Mathematical Physics, LMU, Munich, Germany [link]	10/2023
– Integrability, Dualities and Deformations 2023, Durham University, Durham, UK [link]	07/2023
– Eurostrings 2023, Universidad de Oviedo, Gijon, Spain [link]	04/2023

PARTICIPATION TO POSTGRADUATE SCHOOLS AND WORKSHOPS (*: scheduled, P: poster, T: invited talk, G: gong show)

- Prospects in Theoretical Physics: Celestial Holography, IAS Princeton, NJ, USA [\[link\]](#) (G) 07/2026
- Hidden Structures in Multi-loop Amplitudes, Mainz ITP, Germany [\[link, online\]](#) 04/2026
- Universe+ School: Exploring Positive Geometry, les Houches, France [\[link\]](#) 03/2026
- Single-valued Periods in Scattering Amplitudes, Mainz ITP, Germany [\[link, online\]](#) 01/2026
- Geometry and Combinatorics of Scattering Amplitudes, MPI MiS, Leipzig, Germany [\[link\]](#) 10/2025
- School on Asymptotic Symmetries and Flat Holography, G. Galilei Inst., Florence, Italy [\[link\]](#) (P) 05/2025
- Physics of Machine Learning & ML for physics, Nordita, Stockholm, Sweden [\[link, online\]](#) 01/2025
- Frontiers in Gravity, Niels Bohr Institute, Copenhagen, Denmark [\[link\]](#) 09/2024
- PhD school: Towards Gravity, Nordita, Stockholm, Sweden [\[link, online\]](#) 08/2024
- Celestial Holography Summer School, Perimeter Institute, Waterloo, Canada [\[link, online\]](#) 07/2024
- Celestial 2023, University of Warsaw, Warsaw, Poland [\[link\]](#) 08/2023
- Young Researchers Integrability School & Workshop, Durham University, Durham, UK [\[link\]](#) 07/2023

INVITED TALKS & CONTRIBUTED SEMINARS

- Ambitwistor strings beyond scattering amplitudes
QFT $\times$ String Theory Journal Club, Max Planck Institute for Physics 06/2026
- Wall-crossing structure from quantum phenomena to Feynman integrals
Quantum Field Theory Journal Club, Max Planck Institute for Physics 05/2026
- Resurgence of high-energy string amplitudes
IMPRS Young Scientists Workshop, Castle Aspenstein 07/2026
- String Group Seminar, LMU Munich & Max Planck Institute for Physics 05/2026
- Quantum Geometry Group Seminar, Technical University of Munich 05/2026
- Theory Seminar, Niels Bohr Institute, Copenhagen 04/2026
- Matrix theory reloaded: a BPS road to holography
String Theory Lunch Seminar, LMU Munich & Max Planck Institute for Physics 11/2025
- Boundary Carrollian conformal field theories and open null strings
String Theory Lunch Seminar, LMU Munich & Max Planck Institute for Physics 06/2025
- Spinor-helicity formalism, twistor theory and scattering amplitudes
Seminar on Scattering Amplitudes, LMU Munich 05/2024
- Gauging of discrete higher-form symmetries in non-Abelian Yang-Mills theory
Seminar on Generalized Symmetries in QFT, LMU Munich 12/2023
- Gravitational scattering and covariant phase space methods in gravity
DAMTP Part III Seminar Series, University of Cambridge 03/2023
- Holography and twistor methods in AdS_5
DAMTP Part III Seminar Series, University of Cambridge 12/2022

TECHNICAL SKILLS

- Languages:** French (native speaker); English, German (full work proficiency); Danish (basics)
- IT:** Object-oriented programming (C++, Python), *Mathematica*, MATLAB, $\text{\LaTeX}$ (GitHub+VSCode/Overleaf)

ACADEMIC REFERENCES (thesis advisory committee)

- **Dr. Stephan Stieberger**
Permanent Staff Member & Research Group Leader, String Theory, Max Planck Institute for Physics
- **Prof. Dr. Johannes Henn**
Director & Department Head, Quantum Field Theory, Max Planck Institute for Physics
- **Prof. Dr. Niels Obers**
Professor & Institute Research Deputy Head, Niels Bohr Institute